

NIB 7072: Machine Learning

Course Work: 2025 Batch & Resit

Coventry University / NIBM – MSc Data Science

Question 1: Applied Supervised Machine Learning (40 Marks)

Problem Statement

Select a real-world dataset from a reliable open-source source (Kaggle, UCI Repository, government databases, or industry datasets).

The dataset must:

- Contain at least 15,000 instances.
- Include both numerical and categorical variables.
- Support a supervised learning task (classification or regression).
- Contain realistic challenges such as missing data, class imbalance, noisy attributes, or high dimensionality.

The objective is to design a complete machine learning pipeline, compare different approaches, and provide practical insights.

Tasks

1. Dataset Understanding and Literature Review (10%)

- Describe the dataset source, attributes, size, and business/application context.
- Clearly define the target variable and prediction objective.
- Review at least six recent research papers or industry studies related to similar prediction problems.
- Identify research gaps and explain how your approach addresses them.

2. Data Analysis and Feature Engineering (15%)

- Perform detailed exploratory data analysis using appropriate visualizations.
- Analyze correlations, distributions, and potential data quality issues.
- Handle missing values, outliers, and duplicate records.
- Create at least four new meaningful features.

- Apply suitable feature selection or dimensionality reduction methods (e.g., PCA, feature importance).

3. Model Development and Optimization (25%)

Implement at least five machine learning models from different categories:

- A linear model (Logistic/Linear Regression).
- A distance-based model (KNN).
- A tree-based model (Decision Tree/Random Forest).
- An ensemble boosting model (XGBoost, CatBoost, LightGBM).
- A deep learning model (ANN or another neural network).

Requirements:

- Apply hyperparameter optimization using Grid Search, Random Search, or Bayesian optimization.
- Use cross-validation.
- Justify the selection of algorithms.

4. Performance Evaluation and Model Comparison (15%)

Evaluate models using appropriate metrics:

Classification:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- PR-AUC

Regression:

- MAE
- RMSE
- R^2
- MAPE

Additional requirements:

- Compare all models using tables and visualizations.
- Conduct detailed error analysis.
- Explain why the selected final model is suitable for deployment.

- Apply techniques such as threshold tuning, regularization, or data balancing when required.

5. Explainability and Ethical Considerations (10%)

- Apply explainable AI methods such as SHAP, LIME, or permutation importance.
- Discuss feature influence and decision behavior.
- Analyze fairness, bias, privacy concerns, and ethical implications.

6. Cloud Deployment and MLOps Strategy (25%)

Design a complete cloud-based deployment architecture for the selected machine learning solution. The strategy should include:

- **Cloud Deployment Architecture**
 - Design a deployment solution using cloud platforms such as AWS, Microsoft Azure, Google Cloud Platform (GCP), or other suitable services.
 - Describe the architecture components for model serving, storage, scalability, and security.
- **Model Packaging and Containerization**
 - Package the machine learning model for deployment.
 - Use containerization technologies such as Docker and explain how containers improve portability and scalability.
- **Model Serving and API Development**
 - Design RESTful APIs using frameworks such as FastAPI or Flask for real-time inference.
 - Explain batch inference and real-time inference scenarios where appropriate.
- **MLOps Pipeline and Automation**
 - Design an MLOps workflow covering:
 - Data versioning and model versioning.
 - Experiment tracking.
 - Continuous Integration and Continuous Deployment (CI/CD).
 - Automated testing and deployment pipelines.
- **Monitoring, Maintenance, and Retraining**
 - Propose strategies for:
 - Model performance monitoring.
 - Data drift and concept drift detection.
 - Logging and alerting mechanisms.
 - Automated model retraining and lifecycle management.
- **Deployment Justification**
 - Justify the selected cloud architecture, services, and MLOps practices based on the application requirements.

Question 2: Advanced Time Series Forecasting (30 Marks)

Problem Statement

Choose a time series dataset from domains such as finance, energy, healthcare, retail, transportation, or climate. The dataset should contain sufficient historical observations and demonstrate meaningful temporal patterns such as trend, seasonality, cyclic behavior, or volatility.

Tasks

1. Dataset Understanding and Literature Review (10%)

- Describe the data source, frequency, size, and characteristics.
- Define the forecasting target and prediction horizon.
- Conduct a literature survey of at least five recent studies related to similar forecasting problems.
- Identify limitations of existing approaches and explain the motivation for your selected models.

2. Time Series Exploration and Preprocessing (15%)

- Perform visualization and decomposition of the time series.
- Analyze trends, seasonality, and irregular patterns.
- Detect and handle missing values, anomalies, and outliers.
- Create relevant temporal features such as lag variables, rolling statistics, and seasonal indicators.
- Conduct stationarity tests where appropriate (e.g. ADF test).

3. Model Development: Traditional and Deep Learning Models (35%)

Implement and compare at least three forecasting models

- ARIMA/SARIMA or Facebook Prophet.
- LSTM or GRU-based neural network.
- Machine learning-based forecasting using models such as Random Forest or XGBoost.

Requirements:

- Perform hyperparameter tuning.
- Use appropriate train validation test splitting for time series data.
- Evaluate model performance using MAE, RMSE, MAPE, and other relevant metrics.

4. Volatility Modeling using GARCH (20%)

Implement a GARCH based model for suitable time series data (especially financial or other volatile datasets).

Tasks:

- Explain the concept of conditional heteroscedasticity and volatility clustering.
- Analyze whether the selected time series exhibits changing variance.
- Implement an appropriate GARCH model (e.g., GARCH(1,1)).
- Estimate model parameters and interpret volatility behavior.
- Compare the volatility forecasts with actual observed fluctuations where possible.
- Discuss the strengths and limitations of GARCH models.

5. Model Comparison, Insights, and Critical Reflection (20%)

- Compare forecasting models using quantitative metrics and visualizations.
- Discuss the advantages and limitations of different approaches.
- Interpret practical implications of forecasting results.
- Discuss challenges, assumptions, ethical considerations (if relevant), and future improvements.

Question 3: Optimization using Genetic Algorithms (30 Marks)

Problem Statement

Select a real-world optimization problem such as vehicle routing, scheduling, resource allocation, portfolio optimization, or supply chain optimization.

Tasks

1. Problem Definition and Literature Review (15%)

- Clearly define the optimization objective, decision variables, and constraints.
- Provide a mathematical formulation including:
 - Objective function.
 - Equality and inequality constraints.
 - Integer or binary decision variables where applicable.
- Conduct a literature review of at least five studies related to similar optimization problems and solution methods (Genetic Algorithms, Mixed Integer Programming, heuristics, or metaheuristics).
- Explain the motivation for selecting GA and MIP for the chosen problem.

2. Data Preparation and Problem Analysis (10%)

- Describe the dataset, parameters, and constraints.
- Analyze the complexity of the optimization problem.
- Prepare required representations such as distance matrices, cost matrices, demand matrices, or allocation matrices.
- Explain preprocessing steps and assumptions made.

3. Genetic Algorithm Design and Implementation (30%)

Design and implement a complete Genetic Algorithm including:

- Chromosome representation.
- Population initialization strategy.
- Fitness function design.
- Selection methods (e.g., tournament selection or roulette wheel selection).
- Crossover operators.
- Mutation operators.
- Elitism and stopping criteria.
- Hyperparameter tuning, including:
 - Population size.

- Crossover probability.
- Mutation probability.
- Number of generations.

Include:

- Convergence analysis using fitness improvement plots.
- Evaluation of the best solution obtained.
- Discussion of the computational efficiency and scalability of the GA approach.

4. Mixed Integer Programming (MIP) Formulation and Implementation (25%)

Implement the same optimization problem using a MIP approach.

Tasks:

- Formulate the optimization model mathematically.
- Define objective function, decision variables, and constraints.
- Implement the MIP model using appropriate optimization tools such as:
 - Gurobi.
 - CPLEX.
 - PuLP.
 - OR-Tools.
- Obtain the optimal or near-optimal solution.
- Analyze:
 - Objective function value.
 - Feasibility and constraint satisfaction.
 - Computational time.
 - Scalability for larger problem sizes.

5. Comparative Analysis and Critical Reflection (20%)

Compare the Genetic Algorithm and MIP approaches based on:

- Solution quality and objective value.
- Computational performance and execution time.
- Scalability with increasing problem size.
- Ability to handle complex constraints.
- Flexibility and real-world applicability.

Discuss:

- Advantages and limitations of each method.
- Situations where GA is preferred over MIP and vice versa.
- Possible improvements, including hybrid GA-MIP approaches, parallel optimization, or other metaheuristic methods.
